Solid Waste Management

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Sources of waste generation

- The overall objective of solid-waste management is to minimize the adverse environmental effects caused by the indiscriminate disposal of solid wastes, especially of hazardous wastes.
- To assess the management possibilities it is important to consider:

(1) materials flow in society,

(2) reduction in raw materials usage,

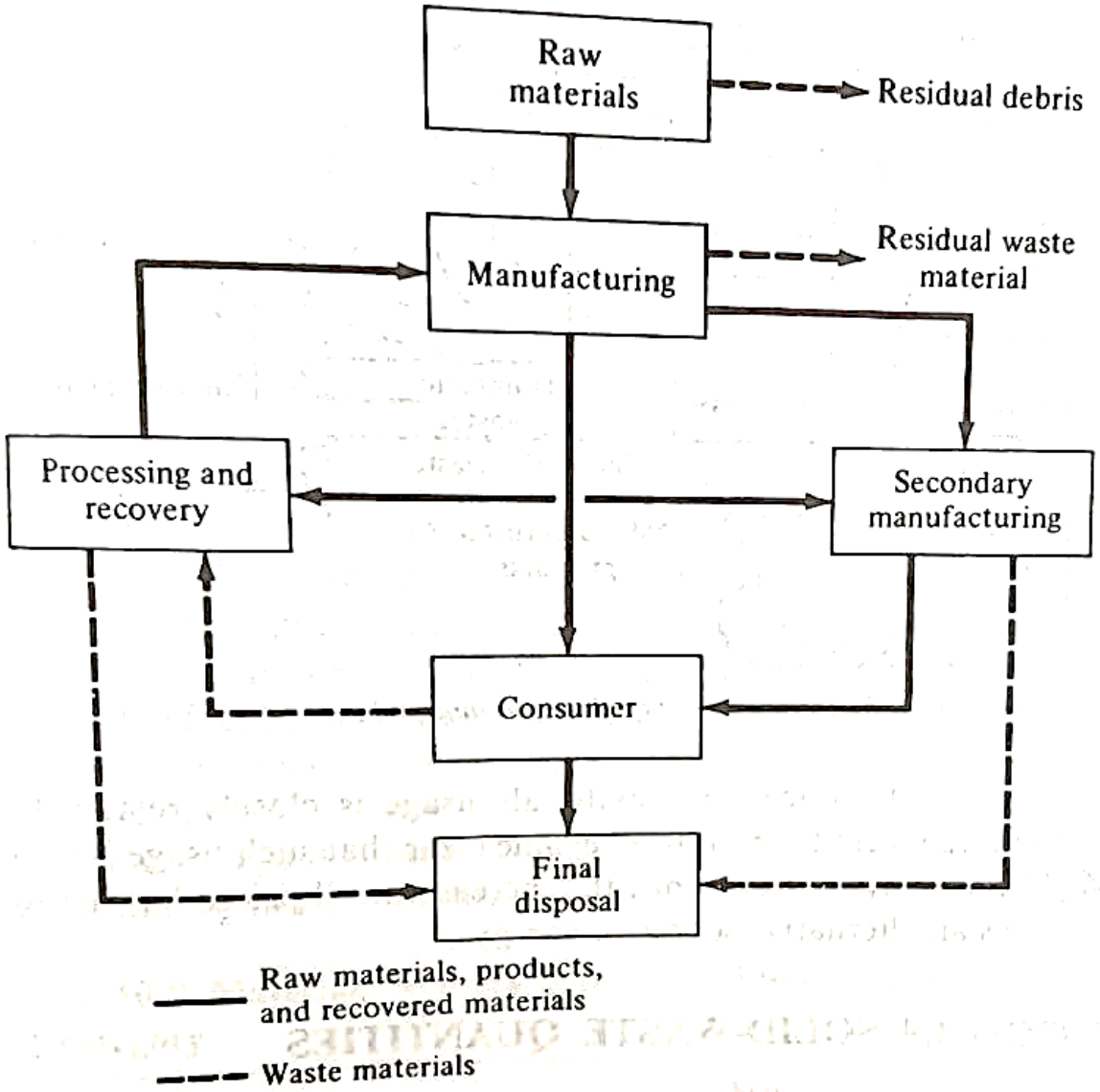
(3) reduction in solid-waste quantities,

(4) reuse of materials,

(5) materials recovery,

(6) energy recovery, and

(7) day-to-day solid-waste management.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

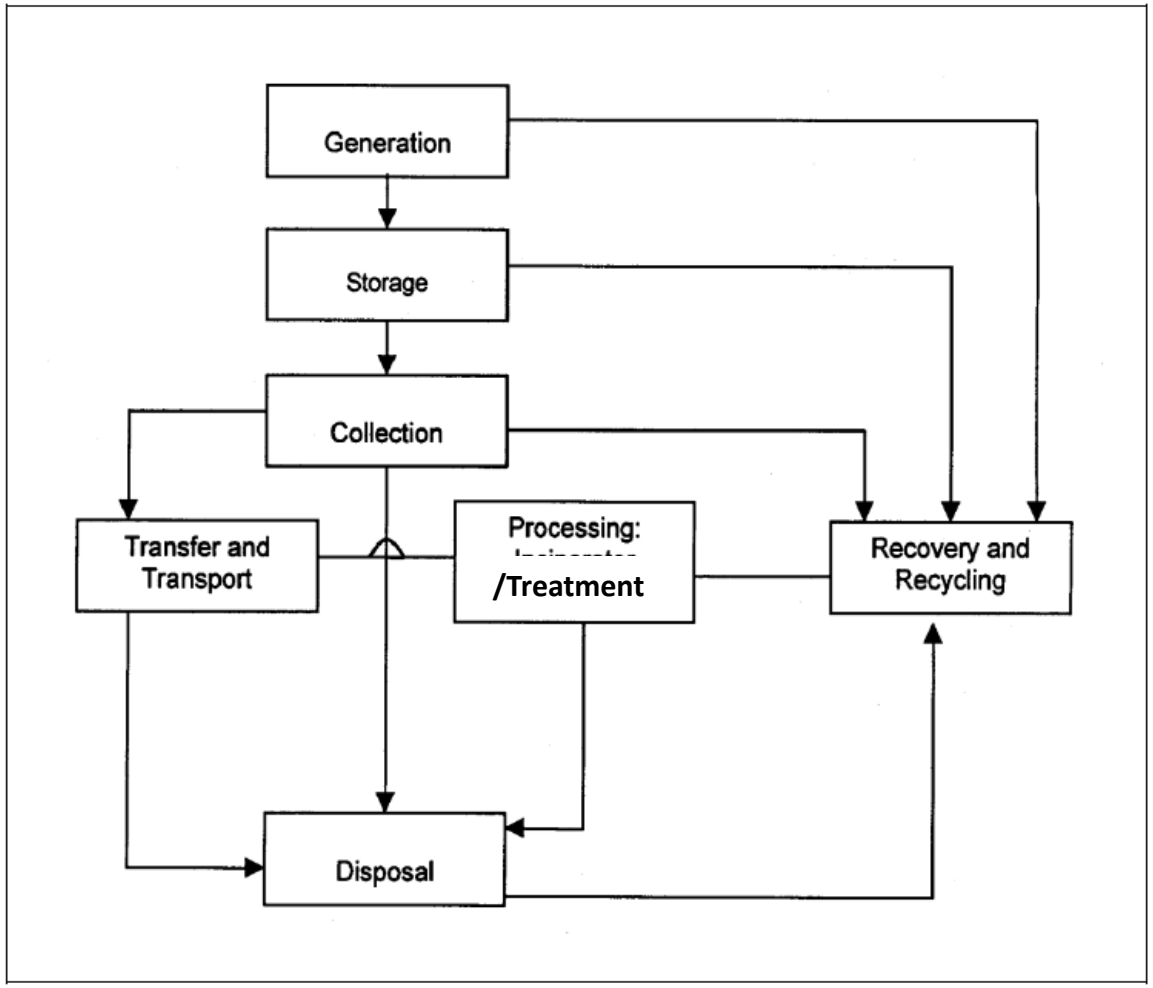
Waste Management Hierarchy

Most preferred- Reduction of waste at source

Least Preferred: Disposal of waste

Source: 1. Environment Protection Agency, Waste management hierarchy, https://www.epa.gov/smm/sustainable-materials-management-non-hazardous-materials-and-waste-management-hierarchy

Functional elements of Solid waste Management



Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Functional elements of Solid waste Management

Functional element	Description
Waste generation	Those activities in which materials are identified as no longer being of value and are either thrown away or gathered together for disposal
On-site handling, storage, and processing	Those activities associated with the handling, storage, and processing of solid wastes at or near the point of generation
Collection	Those activities associated with the gathering of solid wastes and the hauling of wastes after collection to the location where the collection vehicle is emptied
Transfer and transport	Those activities associated with (1) the transfer of wastes from the smaller collection vehicle to the larger transport equipment and (2) the subsequent transport of the wastes, usually over long distance, to the disposal site
Processing and recovery	Those techniques, equipment, and facilities used both to improve the efficiency of the other functional elements and to recover usable materials, conversion products, or energy from solid wastes
Disposal	Those activities associated with ultimate disposal of solid wastes, including those wastes collected and transported directly to a landfill site, semisolid wastes (sludge) from wastewater treatment plants, incinerator residue, compost, or other substances from the various solid-waste processing plants that are of no further use

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Factors affecting Solid Waste Management

- Quantities and characteristics of wastes
- Climate and Season of the year
- Physical characteristics of an urban area
- Financial constraints
- Cultural constraints
- Legislation

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Solid Waste Collection System

The design of an efficient waste collection system requires careful consideration of the type, size and location of containers at the point of generation for storage of wastes until they are collected. While single-family households generally use small containers, residential units, commercial units, institutions and industries require large containers. Smaller containers are usually handled manually whereas the larger, heavier ones require mechanical handling. The containers may fall under either of the following two categories:

- **Stationary containers:** These are used for contents to be transferred to collection vehicles at the site of storage.

- **Hauled containers:** These are used for contents to be directly transferred to a processing plant, transfer station or disposal site for emptying before being returned to the storage site.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Solid Waste Collection Bin

1.1 Cu.m

Stationary Community Bins

4.5 Cu.m

Hauled Community Bins

Movable containers are provided with hoists and tails compatible with lifting mechanism of collection vehicles and such containers have capacities of 1 – 4 Cu.m.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Characteristics of collection container

- The desirable characteristics of a well-designed container are low cost, size, weight, shape, resistance to corrosion, water tightness, strength and durability.
- Containers should not have rough or sharp edges, and preferably have a handle and a wheel to facilitate mobility.
- They should be covered to prevent rainwater from entering (which increases the weight and rate of decomposition of organic materials) into the solid wastes.
- The container body must be strong enough to resist and discourage stray animals and scavengers from ripping it as well as withstand rough handling by the collection crew and mechanical loading equipment.
- Containers should be provided with a lifting bar, compatible with the hoisting mechanism of the vehicle.
- The material used should be light, recyclable, easily molded and the surface must be smooth and resistant to corrosion.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Solid Waste Collection Vehicle

House to House Collection

Hauled container system Vehicle

Image Source: 1. https://www.indiaspend.com/how-indore-became-indias-cleanest-city-and-how-others-can-follow
2. https://images.app.goo.gl/TEBa6dpGQhNHru6Z9

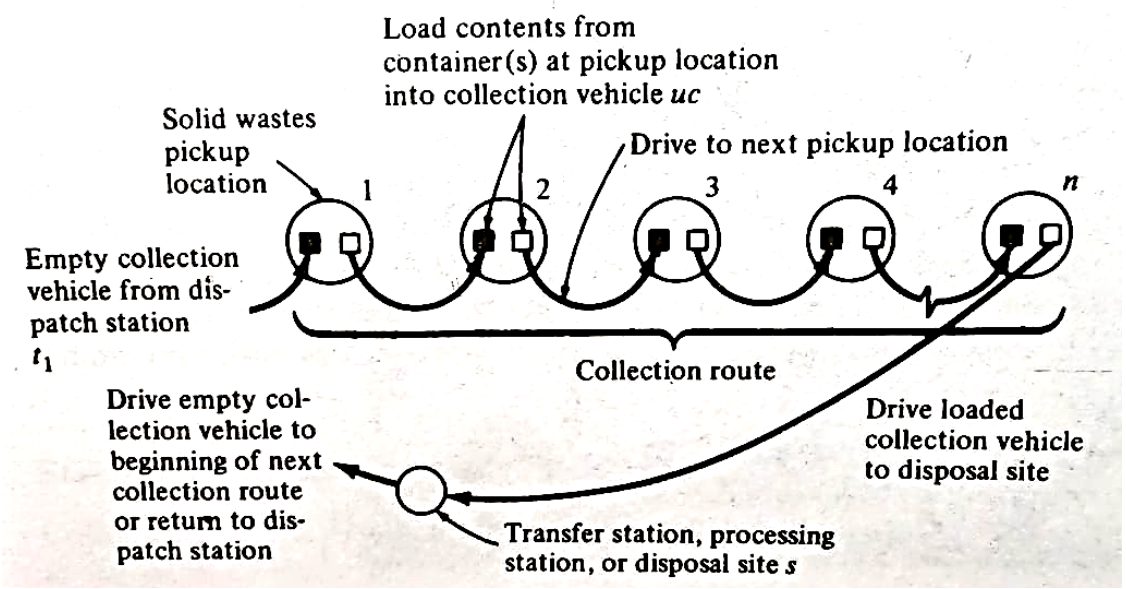
Solid Waste Collection Vehicle

Stationary container system Vehicle

Image Source: 1. https://agartalacity.tripura.gov.in/media-gallery/detail/125/217

Stationary-Container Systems (SCS)

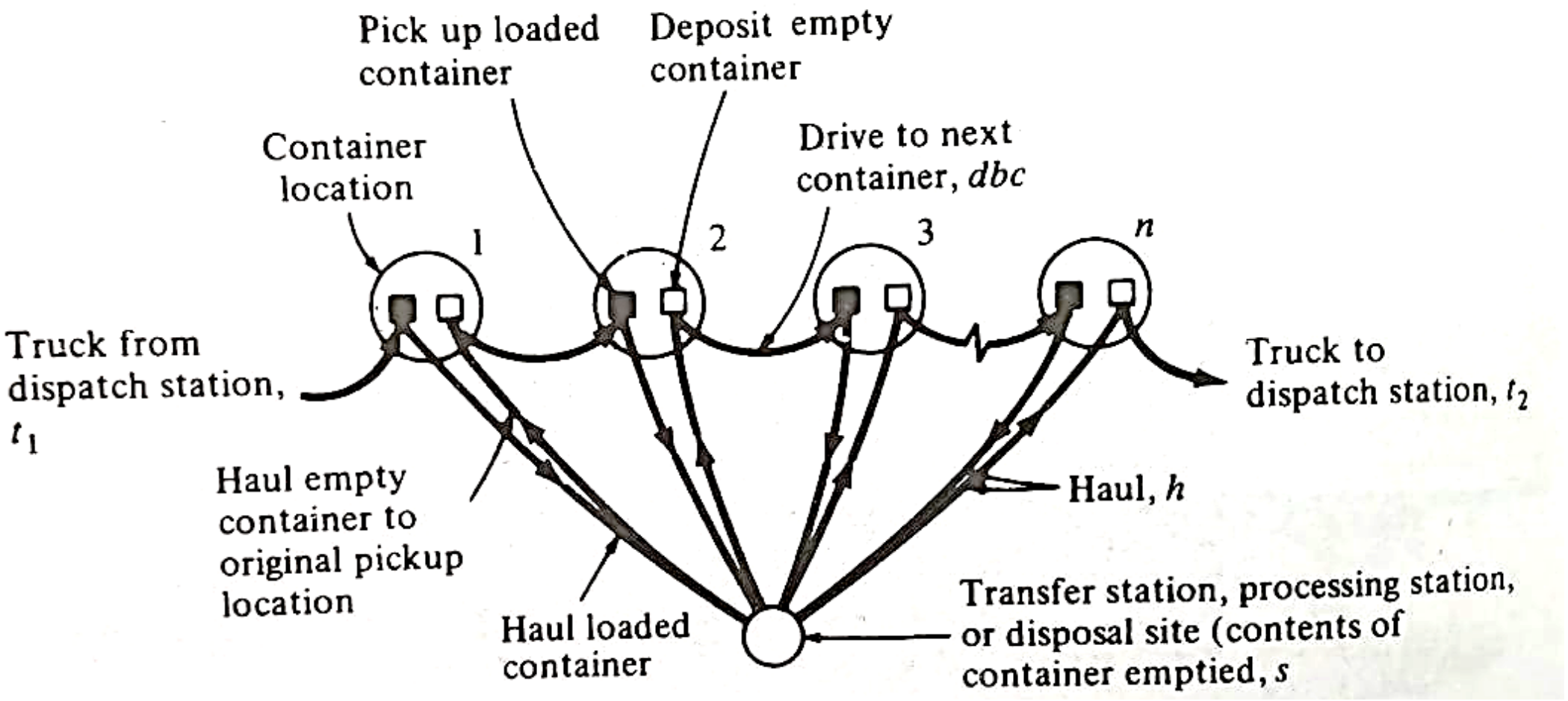
- Collection systems in which the containers used for the storage of wastes remain at the point of waste generation, except when moved for collection are defined as stationary-container systems.



- Trips to the disposal site, transfer station, or processing station are made after the contents of a number of containers have been collected and compacted and the collection vehicle is full.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Hauled-Container Systems (HCS)



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Waste Disposal

Indiscriminate disposal of wastes in many regions is very common, giving rise to such problems as:

- Health hazards (e.g., Residents in the vicinity of wastes inhale dust and smoke when the wastes are burnt; workers and rag pickers come into direct contact with wastes, etc.);
- Pollution due to smoke;
- Pollution from waste leachate and gas;
- Blockage of open drains and sewers.

Clearly, safe disposal of solid wastes is important for safeguarding both public health and the environment.

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Waste Disposal Options

- **Uncontrolled dumping or non-engineered disposal**:

This is the most common method being practised in many parts of the world, and India is no exception. In this method, wastes are dumped at a designated site without any environmental control. They tend to remain there for a long period of time, pose health risks and cause environmental degradation. Due to the adverse health and environmental impact associated with it, the non-engineered disposal is not considered a viable and safe option.

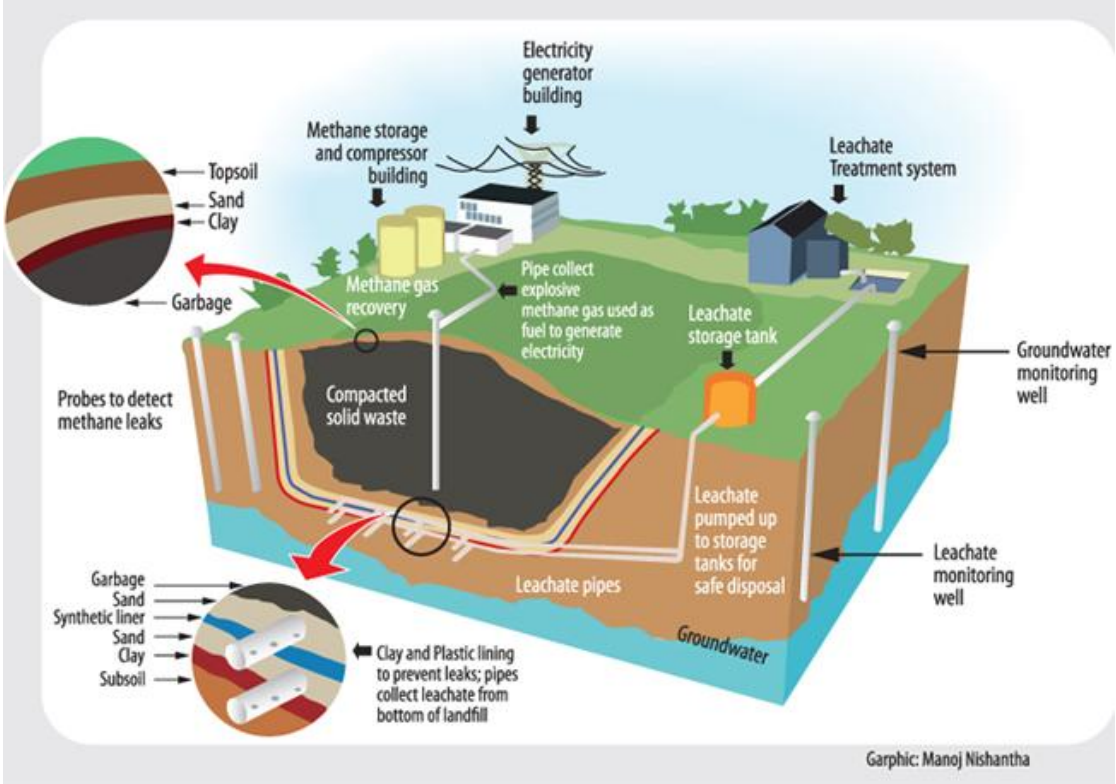
Ghazipur Dumping Yard

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005
2. Image: https://www.downtoearth.org.in/waste/after-fire-outbreaks-ghazipur-landfill-has-a-new-concern-research-finds-its-layers-are-heating-up-95731

Waste Disposal Options

- **Sanitary landfill:**

Unlike the non-engineered disposal, *sanitary landfill is a fully engineered disposal option in that the selected location or wasteland* is carefully engineered in advance before it is pressed into service. Operators of sanitary landfills can minimise the effects of leachate (i.e., polluted water which flows from a landfill) and gas production through proper site selection, preparation and management. This particular option of waste disposal is suitable when the land is available at an affordable price, and adequate workforce and technical resources are available to operate and manage the site.



Sanitary Landfill

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005
2. Image: https://geosrilanka.wordpress.com/tag/sanitary-landfill-sites/

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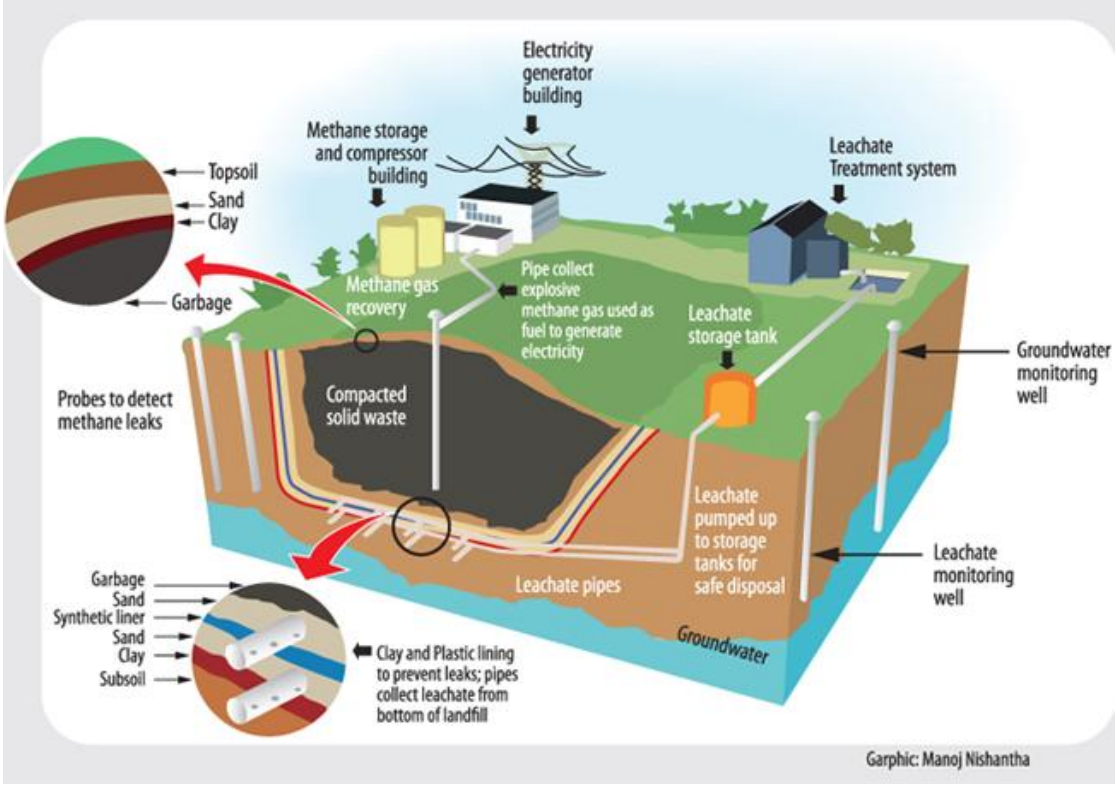

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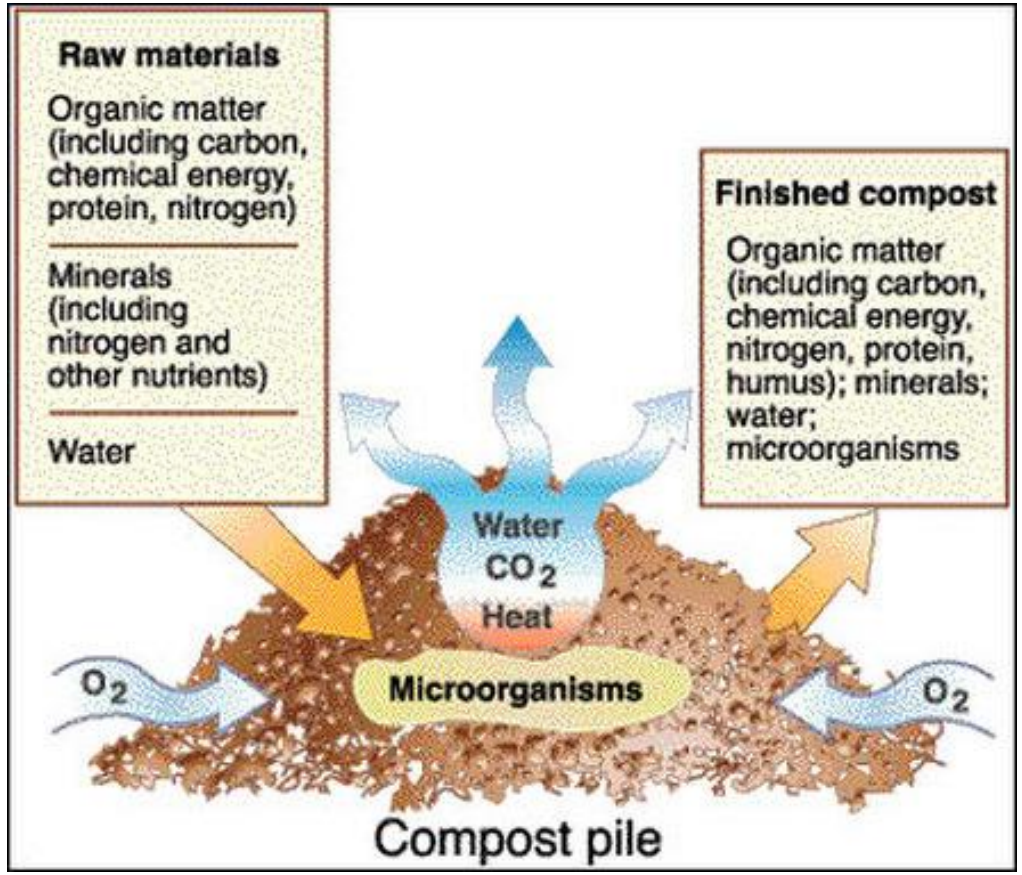
Waste Disposal Options

- **Composting:**

This is a biological process of decomposition in which organisms, under controlled conditions of ventilation, temperature and moisture, convert the organic portion of solid waste into humus-like material. If this process is carried out effectively, what we get as the final product is a stable, odour-free soil conditioner. Generally, the option of composting is considered, when a considerable amount of biodegradable waste is available in the waste stream and there is use or market for composts.



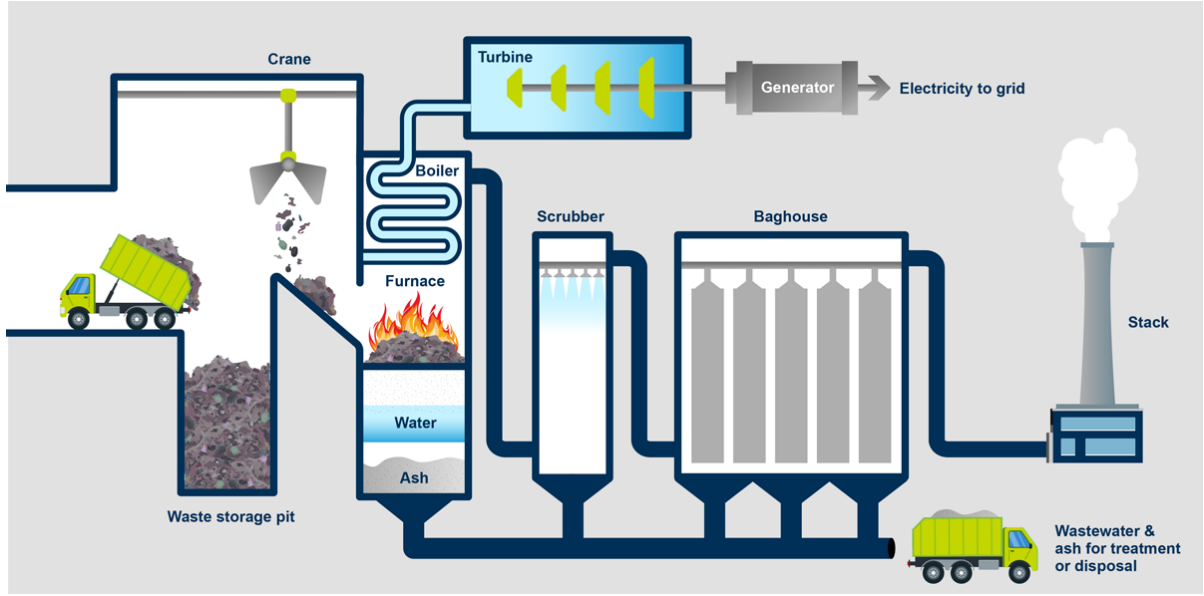
Composting

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005
2. Image: https://fairviewgardencenter.com/blog/composting-basics/

Waste Disposal Options

- **Incineration:**

This refers to the controlled burning of wastes, at a high temperature (roughly 1200 – 1500 degC), which sterilises and stabilises the waste in addition to reducing its volume. In the process, most of the combustible materials (i.e., self-sustaining combustible matter, which saves the energy needed to maintain the combustion) such as paper or plastics get converted into carbon dioxide and ash. Incineration may be used as a disposal option, when land filling is not possible and the waste composition is highly combustible.



Schematic of incineration

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005
2. Image: https://geosrilanka.wordpress.com/tag/sanitary-landfill-sites/

Waste Disposal Options

Disposal Option → ↓Sustainability Indicator	Non-engineered Disposal	Sanitary Landfill	Composting	Incineration
Volume reduction	×	×	×	√
Expensive	×	√	√	√
Long term maintenance	√	√	×	×
By product recovery	×	√	√	√
Adaptability to all wastes	√	√	×	×
Adverse environmental effect	√	√	×	√

Source: 1. T.V.Ramachandran, Municipal Solid Waste management, NPTEL, https://nptel.ac.in/courses/120108005

Thank you

For any questions, please feel free to reach out to me.

Stationary-Container Systems (SCS)

For systems using mechanically self-loading compactors, the time per trip is:

$$T_{SCS} = (P_{SCS} + s + a + bx) \quad (1)$$

T_{SCS}- time per trip for stationary-container systems, h/trip

P_{SCS} = pickup time per trip for stationary-container systems, h/trip

a,b= Haul constant coefficient, h/trip, x = Distance, km

The pickup time for the stationary-container system is given by:

$$P_{SCS} = C_t.uc + (n_p - 1)(dbc) \quad (2)$$

P_{SCS} = pickup time per trip for stationary-container systems, h/trip

C_t= Number of containers emptied per trip, container/trip

uc= average unloading time per container for stationary-container systems, h/container

n_p= Number of container pickup locations per trip, locations/trip

dbc=average time spent driving between container locations, h/location (determined locally)

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Stationary-Container Systems (SCS)

The term n_p-1 accounts for the fact that the number of times the collection vehicle will have be driven between container locations is equal to the number of containers less 1. The number of containers that can be emptied per collection trip is related directly to the volume of the collection vehicle and the compaction ratio that can be achieved. This number is given by:

$$C_t = \frac{vr}{cf} \quad \textbf{(3)}$$

where C_t= number of containers emptied trip, container/trip

v = volume of collection vehicle, m^3 /trip

r = compaction ratio

c = container volume, m^3/container

f = weighted container utilization factor

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Stationary-Container Systems (SCS)

The number of trips required per day is given by:

$$N_d = \frac{V_d}{vr} \quad (4)$$

Where, N_d = number of collection trips required per day, trips/d

V_d = daily waste generation rate, $(m^3)/d$

Where an integer number of trips are to be made each day, the proper combina-tion of trips per day and the size of the vehicle can be determined by using Eq. (11-8) in conjunction with an economic analysis:

$$H = \frac{[(t_1+t_2)+N_d(P_{SCS}+s+a+bx)]}{1-W} \quad (5)$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),

Stationary-Container Systems (SCS)

To determine the required truck volume, two or three different values for N_d are substituted in Eq. (5) and the available pickup times per trip are determined. Then, by trial and error, the truck volume required for each value of N_d is determined using Eqs. (2) and (3). From the available truck sizes, select the ones that most nearly correspond to the computed values. If the available truck sizes are smaller than the required values, compute the actual times per day that will be required using these sizes. The most cost-effective combination then can be selected.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985),